# ALCOHOLS

# LEARNING OBJECTIVES

- By the end of this lecture, students should be able to:
- Describe the mechanism of action of ethanol on the CNS.
- Explain ethanol metabolism.
- Differentiate acute ethanol intoxication from chronic alcohol abuse.
- Outline the treatment of acute ethanol toxicity and alcohol dependence.
- Identify oral manifestations of chronic alcohol abuse.

# WHAT IS ALCOHOL (ETHANOL)?

Ethanol is the type of alcohol found in alcoholic beverages.

Ethanol is a **Central Nervous System (CNS) depressant**.

**Remember**

It **does not stimulate the brain.**

It slows down brain activity.

# MECHANISM OF ACTION OF ETHANOL ON CNS

Ethanol affects different neurotransmitters in the brain.

**1. Increases GABA activity**

GABA is the main inhibitory neurotransmitter.

Normally GABA slows brain activity.

Ethanol increases GABA receptor activity.

**Result:**

Relaxation, Sleepiness, Reduced anxiety, Poor coordination

# 2. DECREASES GLUTAMATE ACTIVITY

Glutamate is the major excitatory neurotransmitter.

It normally keeps the brain active.

Ethanol blocks NMDA receptors.

**Result:**

Memory impairment

Poor learning

Slow thinking

# 3. INCREASES DOPAMINE

Alcohol increases dopamine release in the reward pathway.

**Result:**

Pleasure

Euphoria

Addiction

# 4. INCREASES ENDORPHINS

Endorphins produce a feeling of well-being.

**Result:**

Happiness

Relaxation

# PHARMACOLOGICAL EFFECTS ON CNS

## SMALL DOSES

- Relaxation
- Less anxiety
- Increased talking
- Feeling confident

# MODERATE DOSES

- Slurred speech
- Poor judgment
- Delayed reaction time
- Unsteady walking (ataxia)

# HIGH DOSES

- Confusion
- Vomiting
- Loss of consciousness
- Respiratory depression
- Coma
- Death

# METABOLISM OF ETHANOL

- After ingestion, ethanol is rapidly and completely absorbed, distributed to most body tissues, and its volume of distribution is ~0.5–0.7 L/kg.

- Most ethanol is metabolized in the liver.
- Two main enzyme systems metabolize ethanol to acetaldehyde: **Alcohol dehydrogenase (ADH)** and the **microsomal ethanol-oxidizing system** (MEOS).

# STEP 1(ALCOHOL DEHYDROGENASE {ADH})

**Location:**

Liver cytoplasm

**Reaction:**

Ethanol

↓

Acetaldehyde

This is the first enzyme

# STEP 2
# {ALDEHYDE DEHYDROGENASE (ALDH)}

**Location:**

Mitochondria

**Reaction:**

Acetaldehyde

↓

Acetate

Acetate is less toxic.

# STEP 3

Acetate

↓

Carbon dioxide + Water + Energy

**Acetaldehyde is highly toxic.**

It causes:

Flushing ,Headache

Nausea ,Vomiting

# MICROSOMAL ETHANOL-OXIDIZING SYSTEM (MEOS)

**What is MEOS?**

MEOS is an **alternative pathway** that helps the liver break down alcohol.

🧠 **Main enzyme: CYP2E1** (a cytochrome P450 enzyme)

# WHEN DOES MEOS BECOME ACTIVE?

Normally, alcohol is metabolized mainly by **Alcohol Dehydrogenase (ADH)**.

When a person drinks **large amounts of alcohol (blood alcohol >100 mg/dL)** or drinks **frequently**, the **MEOS pathway becomes active**.

# WHAT HAPPENS IN CHRONIC ALCOHOL USE?

Chronic alcohol consumption:

⬆️ Increases (induces) **CYP2E1 enzyme**

↓

⬆️ Increases MEOS activity

↓

Alcohol is broken down faster

↓

**Tolerance develops** (the person needs more alcohol to get the same effect)

↓

More toxic free radicals are produced

↓

**Liver damage (hepatotoxicity)**

# ADH vs MEOS Pathway

| 🟦 Alcohol Dehydrogenase (ADH) | 🟧 MEOS (CYP2E1) |
|---|---|
| ⭐ Main pathway | 🔄 Alternative pathway |
| 🍷 Normal alcohol intake | 🍺 Heavy/chronic alcohol use |
| 🧪 Enzyme: ADH | 🧬 Enzyme: CYP2E1 |
| ⚡ Requires $NAD^+$ | ⚡ Requires NADPH + $O_2$ |
| ➡ Produces acetaldehyde | ➡ Also produces acetaldehyde |
| ✔ No tolerance | ✔ Causes tolerance |
| ✔ Minimal liver injury | ⚠️ Free radicals → hepatotoxicity |
| 👤 Predominant pathway | 🍾 Active in chronic alcoholics |

# ACUTE ETHANOL INTOXICATION

- Alcohol consumed in a large amount over a short period.

**Clinical Features**

**MILD**

- Euphoria
- Talkative
- Reduced inhibition

| Stage | Clinical Features | Key Message |
| --- | --- | --- |
| 🟢 Mild | • Euphoria<br>• Talkative<br>• Reduced inhibition | Patient is awake and relaxed |
| 🟡 Moderate | • Slurred speech<br>• Blurred vision<br>• Poor coordination<br>• Ataxia | Impaired judgment and balance |
| 🔴 Severe | • Vomiting<br>• Hypothermia<br>• Respiratory depression<br>• Coma<br>• Death | Medical emergency |
| 💡 Memory Tip | Mild → Moderate → Severe | Symptoms worsen as blood alcohol level increases |
| ⚠️ Note | Respiratory depression is the major cause of death | Requires immediate supportive care |

# Chronic Alcohol Abuse – Systemic Effects

| System | Major Effects |
|---|---|
| Brain | • Dependence<br>• Memory loss<br>• Peripheral neuropathy<br>• Dementia |
| Liver | • Fatty liver<br>• Alcoholic hepatitis<br>• Cirrhosis |
| Heart | • Cardiomyopathy<br>• Hypertension<br>• Arrhythmias |
| Pancreas | • Acute & Chronic pancreatitis |
| GI Tract | • Gastritis<br>• Peptic ulcer<br>• GI bleeding |
| Nutrition | • Vitamin B1 deficiency<br>• Wernicke encephalopathy<br>• Korsakoff syndrome |

# WERNICKE–KORSAKOFF SYNDROME

- Due to thiamine deficiency, seen in chronic alcoholics.
- **Wernicke's triad:** confusion, ataxia, ophthalmoplegia.
- **Korsakoff's syndrome:** anterograde amnesia, confabulation.
- **Treatment:** administer thiamine before glucose.

# Management of Acute Alcohol Poisoning

Patient with Alcohol Poisoning

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ABC Assessment (Airway • Breathing • Circulation)

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Oxygen + IV Fluids + Monitoring

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Check Blood Glucose

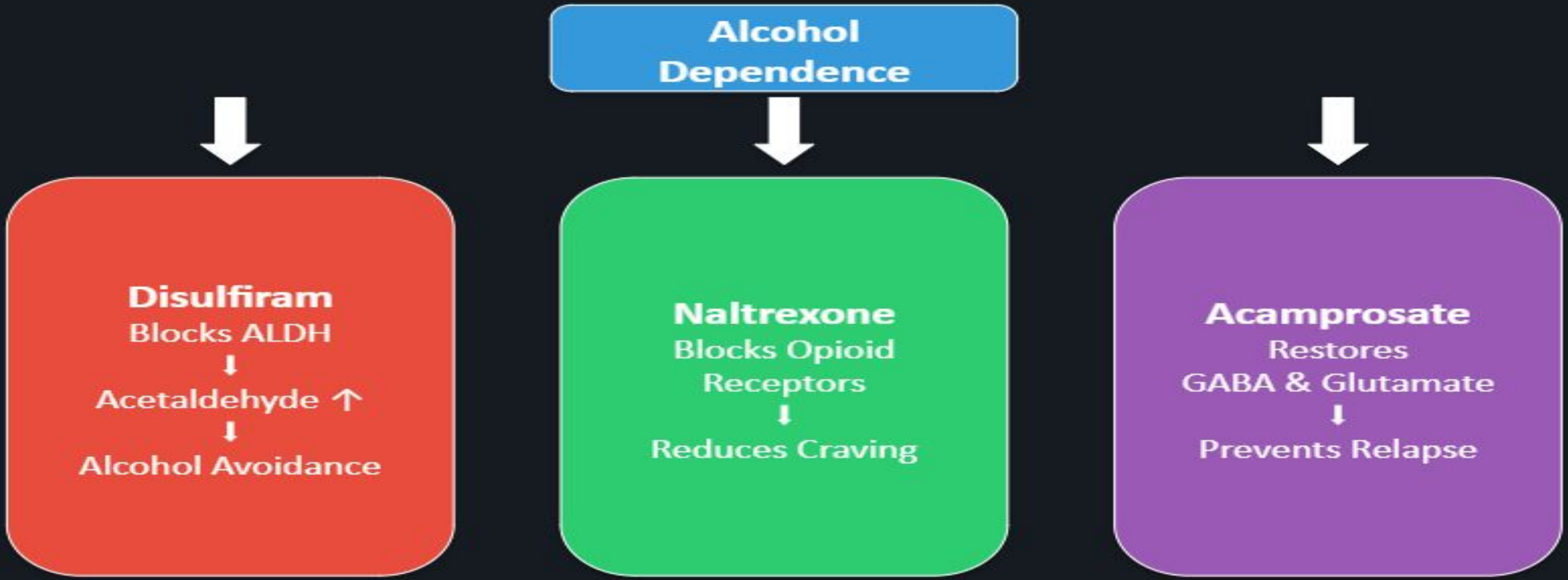
↓

Give THIAMINE BEFORE GLUCOSE

↓

Severe Cases Intubation & Ventilation

# Drugs Used in Alcohol Dependence



# DRUGS USED IN ALCOHOL DEPENDENCE

**3 MAIN DRUGS – 3 DIFFERENT ACTIONS**

## 1 DISULFIRAM
(Alcohol = Sickness)

### HOW IT WORKS

Alcohol → Acetaldehyde (Toxic) → ALDH BLOCKED → Acetaldehyde builds up

### WHAT HAPPENS WHEN ALCOHOL IS DRUNK?

- Flushing (red face)
- Nausea & Vomiting
- Headache
- Palpitations
- Severe discomfort

↓

**PATIENT LEARNS:**
"Drinking alcohol causes severe discomfort."
→ Alcohol Avoidance

## 2 NALTREXONE
(Alcohol = No Pleasure)

### HOW IT WORKS

Alcohol → Opioid receptors in Reward Centre → Naltrexone blocks receptors → Less dopamine released

### WHAT HAPPENS?

- ↓ Dopamine
- Less pleasure
- Craving decreases
- Relapse risk decreases

↓

**PATIENT THINKS:**
"I drink alcohol but don't feel the pleasure."
→ Craving Decreases

## 3 ACAMPROSATE
(A Calm Brain)

### HOW IT WORKS


After stopping alcohol (Brain disturbed) → GABA (Calm) / Glutamate (Excitement) → Acamprosate restores balance

### RESULT

- GABA & Glutamate balanced
- Anxiety, irritability, insomnia improve
- Craving decreases
- Prevents relapse

↓

**BRAIN BECOMES NORMAL AGAIN**
→ Relapse Prevented

## EASY COMPARISON

| DRUG | EASY ACTION | RESULT |
|---|---|---|
| Disulfiram | Makes alcohol cause sickness (Blocks ALDH) | Alcohol avoidance |
| Naltrexone | Blocks opioid receptors (Blocks reward) | Craving decreases |
| Acamprosate | Restores brain chemicals (GABA & Glutamate) | Prevents relapse |

## ONE-LINE MEMORY TRICK

- Disulfiram → Alcohol = Sickness
- Naltrexone → Alcohol = No Pleasure
- Acamprosate → A Calm Brain

## KEY TAKE HOME MESSAGE

These drugs help the patient stop drinking, reduce craving, and prevent relapse through different mechanisms.

**★ REMEMBER:** Different Drug – Different Mechanism – Same Goal → Recovery

# ORAL MANIFESTATIONS OF CHRONIC ALCOHOL ABUSE

## 1. Xerostomia (Dry Mouth)

- Alcohol decreases saliva production.

**Problems:**

- Difficulty swallowing
- Bad breath
- Increased dental caries

# 2. PERIODONTAL DISEASE

Poor oral hygiene

Smoking

Reduced immunity

↓

Gingivitis

↓

Periodontitis

↓

Tooth loss

# DENTAL CARIES

Dry mouth increases cavities.

**Oral Candidiasis**

White fungal patches in the mouth.

**Halitosis**

Bad breath.

# DELAYED WOUND HEALING

After extraction or oral surgery.

**Increased Risk of Oral Cancer**

Especially when alcohol is combined with smoking.

# METHANOL TOXICITY

- Methanol (wood alcohol) is metabolized to formaldehyde and formic acid, leading to severe metabolic acidosis, retinal damage, and blindness.
- Treatment: fomepizole (ADH inhibitor) or ethanol to competitively inhibit methanol metabolism.

# ETHYLENE GLYCOL TOXICITY

- Found in antifreeze, metabolized to oxalic acid causing renal failure and acidosis.
- Treatment: fomepizole and correction of acidosis. Oxalate crystals in urine are characteristic.

# DRUGS USED IN ALCOHOL DEPENDENCE

- Disulfiram: Inhibits aldehyde dehydrogenase, causes aversive reaction to alcohol.
- Naltrexone: Opioid antagonist that reduces craving.
- Acamprosate: NMDA receptor modulator that helps maintain abstinence.